

Advanced Configuration of Therefore™ Web Access and Therefore™ Portal

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1. Introduction

The purpose of this white paper is to provide you with information on advanced configuration and customization possibilities for Therefore™ Web Access and the Therefore™ Portal.

2. Customization

2.1 What are Web Access customizations

Therefore customization should enable users to extend the Therefore™ Web Access using web technologies like HTML, CSS and JavaScript.

Customizing the Therefore™ Web Access means that a user can alter the UI to meet their practical requirements, this includes:

- Adding new functionalities – e.g. add new button to the RibbonBar, which can retrieve the selected items from the hit-list and execute valid XML Web Service call by using the current user (with token) and much more. If the token does not exist, Windows credentials can be used.
- Removing existing functionalities – e.g. hiding existing buttons (along with their functions) from the RibbonBar.
- Changing functionalities – e.g. inputting field on a category dialog can have special functionality, which does not exist externally.
- Altering the aesthetics of the User Interface – e.g. amending the layout, color-scheme and appearance of the Web Access' UI.

2.2 How can we customize Web Access?

Configuration of the Web Access' customization should be done via the manifest XML file, which specifies settings and capabilities of the add-in.

We provide a framework for Web Access customization that is installed along with Web Access, and must be used when modifying the UI and functionality.

Here are the instructions for customizing Therefore™ Web Access based on the initiatives mentioned in Chapter 1:

- Adding new functionalities:
 1. Create a manifest XML file that holds the definition of the customization - this should define what the user would like to customize and how they intend to implement the customization.
 2. Create a function file (HTML), which must also be defined on the manifest file – this can be on the same server where the Therefore™ client has been installed. Alternatively, it can also be installed on a different server; however, the communication protocols must be consistent (i.e. HTTP → HTTP, → HTTPS). The function file must use the Therefore™ customization framework, which is installed along with the Web Access (*therefore.js* or the debug version of it *therefore.debug.js*).
 3. Adding new items to the RibbonBar with specific images for the buttons is optional.
 4. Once customization has been finalized, an administrator/operator/add-in manager has to upload it via Web Access (i.e. Therefore™ Web Navigator -> File menu -> Settings -> Manage Add-ins).
- Removing existing functionalities:

1. Create a manifest XML file, which holds the definition of the customization - this should define what the user would like to customize and how they intend to implement customization.
2. To hide the button from the RibbonBar, add the button IDs to the "hide list" in the XML file (<HideList>).

- Changing functionalities:

1. As an example, the calculated fields where a change has been made to the default behavior of the input fields and where new ones were added to it. This would require a function file, without one it is not possible to achieve. On the manifest XML file you can define which input event(s) you would like to customize and in which function you would like this to be applied.
2. One add-in can add only one group to an existing tab.
3. Only one custom tab can be added to the RibbonBar and a maximum of six groups to the custom tab.

- Altering the aesthetics of the User Interface:

1. This can be done by removing buttons from the RibbonBar or customizing the index data dialog.
2. As an example, the user can change the background color to their desired preference.

2.3 How to develop your own customization?

By using the Therefore™ Web Access framework, you can acquire and set values of the controls. This is always limited by the current version of the framework (i.e. not all controls will be available and neither will all properties).

Customization is heavily web technology based (HTML, JavaScript, etc.), and hence, connections between the Web Client and the customization provider must be consistent:

- HTTP -> HTTP
- HTTPS -> HTTPS

This means that these protocols cannot be different (i.e. HTTP -> HTTPS or HTTPS -> HTTP), otherwise, the intended customization will not be achieved.

The Web Access should have direct access to the server with the user's retained data, the server's name should follow the backslashes then followed by the pathway as shown in the below example:

[\\\\$SERVER\\$c\\$\inetpub\wwwroot\TWA\Client\bin\XSD\](\\$SERVER$c$\inetpub\wwwroot\TWA\Client\bin\XSD\)

The XSD files can be used as the manifest file, assuming that a toolkit has already been developed in order for it to be understood. This can validate the manifest file on the fly, and can offer valid parameters and options. It is imperative for the developer to ensure that the XSD files are activated.

Templates are available for download in the following link:

<https://extranet.therefore.net/downloads/?f=3145#cat11>

2.4 XML Manifest File

An XML manifest file can be obtained by:

1. Using templates
2. Opening the Xml Templates with Xml Validation Tools. For instance:
 - a) Visual Studio: right click on xml File and select Open with "Visual Studio" Then on XML select Schemas... Add XSD Files located under "C:\inetpub\wwwroot\TWA\Client\bin\XSD"
 - b) Any Other XML Validation Tool.

2.4.1 ThereforeAddIn

This is the root element in the manifest of a ThereforeAddIn.

Example:

```
<?xml version="1.0" encoding="utf-8" ?>
<the:ThereforeAddIn
  xmlns:the="http://schemas.therefore.net/thereforeaddinmanifest/0.1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:bt="http://schemas.therefore.net/thereforebasictypes/0.1">
  <Id>73577357-abcd-7357-ffff-735773570004</Id>
  <Version>0.1</Version>
  <ProviderName>Therefore</ProviderName>
  <DefaultLocale>en-US</DefaultLocale>
  <DisplayName DefaultValue="Test XmlWebSerivce">
    <Override Locale="de-DE" Value="asfasda"/>
  </DisplayName>
  <Description DefaultValue="The Description" />
  <IconUrl DefaultValue="/TWA/Client/Images/therefore_32.png" />
  <Hosts>
    ...
  </Hosts>
</the:ThereforeAddIn>
```

Attributes of ThereforeAddIn:

xmlns:the	Defines the ThereforeAddIn manifest namespace and schema version, and should be set to: "http://schemas.therefore.net/thereforeaddinmanifest/0.1"
xmlns:xsi	Defines the XMLSchema instance, and should be set to: "http://www.w3.org/2001/XMLSchema-instance"
xmlns:bt	Defines the ThereforeAddIn type, and should be set to: "http://schemas.therefore.net/thereforebasictypes/0.1"

Mandatory Elements of ThereforeAddIn:

Id	Specifies the unique ID of your ThereforeAddIn as a UUID
Version	Specifies the version of the ThereforeAddIn
ProviderName	States the name of the developer of the ThereforeAddIn within a string of no more than 125 characters
DefaultLocale	States the default culture name of the locale used by strings in the ThereforeAddIn.
DisplayName	States the name the ThereforeAddIn as a string up to 125 characters long. This element is of Type LocaleAwareSetting.
Description	States the description of the ThereforeAddIn as a string of no longer than 250 characters. This element is of Type LocaleAwareSetting.

Optional Elements of ThereforeAddIn:

IconUrl	This element is of Type LocaleAwareSetting.
Groups	Specifies on which UserGroups should the AddIn be applied.
Hosts	Specifies the Office client application where the app for Office will activate.
Resources	Specifies a collection of resources referenced by other elements of the manifest where resource references are supported.

2.4.2 Groups

Specifies on which UserGroups should the AddIn be applied.

To look up the User or Group ID, open Solution Designer and go to Design > Users and Groups

Example:

```
<Groups>
  <GroupId>12</GroupId>
</Groups>
```

Mandatory Elements of Groups:

GroupId	User or Group ID
---------	------------------

2.4.3 Hosts

Specifies the Therefore client application where the customization should apply.

Example:

```
<Hosts>
  <Host xsi:type="the:Navigator">
    ...
  </Host>
</Hosts>
```

2.4.3.1 Host

Specifies the Therefore client application where the Customization should apply.

For now, there is only one Host Type: "Navigator".

Mandatory Elements of Host:

Type of FormFactor	Specifies the list of settings for the Therefore add-in when activated for a given form factor of the Therefore client application. For now there is only "DesktopFormFactor"
--------------------	--

Example:

```
<Hosts>
  <Host xsi:type="the:Navigator">
    <DesktopFormFactor>
      ...
    </DesktopFormFactor>
  </Host>
</Hosts>
```

2.4.4 FormFactor

Specifies the list of settings for the Therefore add-in when activated for a given form factor of the Therefore client application. The only type that is currently available type is "DesktopFormFactor".

Mandatory Elements of FormFactor:

FunctionFile	Specifies the source code file for the Therefore add-in, containing operations it exposes via its extension points, to use when loaded in this form factor.
ExtensionPoint	Specifies a location in a Therefore client application where the Therefore add-in registers itself to perform operations

Example:

```
<Host xsi:type="the:Navigator">
  <DesktopFormFactor>
    <FunctionFile resid="Therefore.FunctionFile.Url"/>
    <ExtensionPoint xsi:type="the:CategoryIndexDataDialog" CategoryNo="12">
      ...
    </ExtensionPoint>
    <ExtensionPoint xsi:type="the:NavigatorRibbonBar">
      ...
    </ExtensionPoint>
  </DesktopFormFactor>
</Host>
```

2.4.4.1 FunctionFile

Specifies the source code file for operations that an add-in exposes through add-in “Action”, which executes a JavaScript function.

Mandatory Attributes of FunctionFile:

resid	Specifies a reference to a URL resource defined under the Resources element of the manifest.
-------	--

2.4.4.2 ExtensionPoints

Specifies a location in the Therefore client application where the Therefore add-in exposes functionality.

Mandatory Attributes of ExtensionPoints:

xsi:type	Specifies the type of the location in the Therefore client application. Possible types: • IndexDataDialog • CategoryIndexDataDialog • CaseDefinitionIndexDataDialog • RibbonBarExtensionPoint • NavigatorRibbonBar • ViewerRibbonBar • CaseViewerRibbonBar • TabStrip • Hitlist
----------	---

2.4.4.2.1 IndexDataDialog

Base Type for “CategoryIndexDataDialog” and “CaseDefinitionIndexDataDialog”.

Optional Attributes of IndexDataDialog:

OnQuery	Specify if customization should be applied on Query Mode
OnArchive	Specify if customization should be applied on Archive Mode
OnUpdate	Specify if customization should be applied on Update Mode

Remarks on the optional attributes:

- By default customization will be applied on all modes.
- As soon one of these attributes are set, customization will only be applied on modes that are set to "true".
- All the above modes set to "false" will not be supported; in this instance, all will be automatically set to "true".

Optional Attributes of EventTriggers:

EventTriggers	Specifies a collection of Events.
---------------	-----------------------------------

Example:

```
<ExtensionPoint xsi:type="the:CategoryIndexDataDialog" CategoryNo="12"
OnQuery="true">
  <EventTriggers>
    <Event xsi:type="the:OnInit">
      <Action xsi:type="the:ExecuteFunction">
        <FunctionName>aasa</FunctionName>
      </Action>
    </Event>
  </EventTriggers>
</ExtensionPoint>
```

2.4.4.2.2 CategoryIndexDataDialog

This is derived from IndexDataDialog.

Mandatory Attributes of CategoryIndexDataDialog:

CategoryNo	Specifies the Category Number, where the customization should be applied.
------------	---

Example:

```
<ExtensionPoint xsi:type="the:CategoryIndexDataDialog" CategoryNo="6"
OnArchive="true">
  <EventTriggers>
    ...
  </EventTriggers>
</ExtensionPoint>
```

2.4.4.2.3 CaseDefinitionIndexDataDialog

Derived from IndexDataDialog.

Mandatory Attribute of CategoryDefinitionIndexDataDialogs:

CaseDefNo	Specifies the Case Definition Number, where the customization should be applied.
-----------	--

Example:

```
<ExtensionPoint xsi:type="the:CategoryIndexDataDialog" CaseDefNo="7"
OnArchive="true">
  <EventTriggers>
    ...
  </EventTriggers>
</ExtensionPoint>
```

2.4.4.2.4 RibbonBarExtensionPoint

Base Type for "NavigatorRibbonBar", "ViewerRibbonBar" and "CaseViewerRibbonBar".

Optional Elements of RibbonBarExtensionPoint:

HideList	Specifies a collection of Controls which should be hidden.
Tab	Specify which existing Tab should be customized.
CustomTab	Specify a new custom Tab.

Example:

```
<ExtensionPoint xsi:type="the:NavigatorRibbonBar">
  <HideList>
    ...
  </HideList>
  <Tab id="theRibbonTabHOME">
    ...
  </Tab>
  <CustomTab id="Therefore.CustomTab">
    ...
  </CustomTab>
</ExtensionPoint>
```

HideList

Specifies a collection of controls that should be hidden.

Mandatory Elements of HideList:

ControlId	Specify the RibbonBar Control which should be hidden.
-----------	---

Example:

```
<HideList>
  <!-- RibbonBar ApplicationMenu (Context Menu) -->
  <!--<ControlId>appMenuItemNEW</ControlId>-->
    <!--<ControlId>appMenuItemNEWDOC</ControlId>-->
    <!--<ControlId>appMenuItemNEWCASE</ControlId>-->
  <!--<ControlId>appMenuItemCOPY</ControlId>-->
  <!--<ControlId>appMenuItemPRINT</ControlId>-->
  <!--<ControlId>appMenuItemRECYCLEBIN</ControlId>-->
  <!--<ControlId>appMenuItemREFRESHTREEVIEW</ControlId>-->
  <!--<ControlId>appMenuItemADMINISTRATIVETOOLS</ControlId>-->
  <!--<ControlId>appMenuItemOUTOFOFFICE</ControlId>-->
  <!--<ControlId>appMenuItemDOWNLOADGADGET</ControlId>-->

  <!-- RibbonBar -->

  <!-- Tab HOME -->
  <!--<ControlId>theRibbonTabHOME</ControlId>-->
    <!--<ControlId>theRibbonGroupSEARCH</ControlId>-->
      <!--<ControlId>btnSEARCH</ControlId>-->
        <ControlId>btnSearchNEWSEARCH</ControlId>
        <!--<ControlId>btnSearchNEWCROSSCATEGORYSEARCH</ControlId>-->
      <!--<ControlId>btnOpenSearch</ControlId>-->
    <!--<ControlId>theRibbonGroupFULLTEXT</ControlId>-->
      <!--<ControlId>txtbxQuickSearch</ControlId>-->
      <!--<ControlId>cbxDocumentContent</ControlId>-->
      <!--<ControlId>cbxIndexData</ControlId>-->
      <!--<ControlId>btnFullTextADVANCEDSEARCH</ControlId>-->
    <!--<ControlId>theRibbonGroupHITLIST</ControlId>-->
      <!--<ControlId>btnHitlistMODE</ControlId>-->
      <!--<ControlId>btnHitlistREFRESH</ControlId>-->
      <!--<ControlId>btnHitlistPRINT</ControlId>-->
      <!--<ControlId>btnHitlistCOPY</ControlId>-->
    <!--<ControlId>theRibbonGroupDOCUMENTS</ControlId>-->
      <!--<ControlId>btnDocumentsTEMPLATES</ControlId>-->
      <!--<ControlId>btnDocumentsVIEW</ControlId>-->
      <!--<ControlId>btnDocumentsPRINT</ControlId>-->
      <!--<ControlId>btnDocumentsINDEXDATA</ControlId>-->
      <!--<ControlId>btnDocumentsEXPORT</ControlId>-->
        <!--<ControlId>btnDocumentsExportEXPORT</ControlId>-->
        <!--<ControlId>btnDocumentsExportSEND</ControlId>-->
      <!--<ControlId>btnDocumentsLINKS</ControlId>-->
      <!--<ControlId>btnDocumentsSECURITY</ControlId>-->
      <!--<ControlId>btnDocumentsHISTORY</ControlId>-->
      <!--<ControlId>btnDocumentsDELETE</ControlId>-->
    <!--<ControlId>theRibbonGroupPROCESS</ControlId>-->
      <!--<ControlId>btnProcessNEWTASK</ControlId>-->
      <!--<ControlId>btnProcessSTART</ControlId>-->
      <!--<ControlId>btnProcessLINKTO</ControlId>-->
    <!--Custom Tab with Custom Button-->
```

```

<!--<ControlId>the-Therefore.FunctionButton1</ControlId>-->

<!-- Tab WORKFLOW -->
<!--<ControlId>theRibbonTabWORKFLOW</ControlId>-->
  <!--<ControlId>theRibbonGroupVIEWS</ControlId>-->
    <!--<ControlId>btnViewsMYINSTANCES</ControlId>-->
      <!--<ControlId>btnViewsALLINSTANCES</ControlId>-->
        <!--*All Instances-->
        <!--*All Running-->
        <!--*All Finished-->
        <!--*All Errors-->
      <!--<ControlId>btnViewsCONFIGURE</ControlId>-->
        <!--*My Instances-->
        <!--*All Instances-->
        <!--*All Running-->
        <!--*All Finished-->
        <!--*All Errors-->
    <!--<ControlId>theRibbonGroupINSTANCES</ControlId>-->
      <!--<ControlId>btnInstancesOPEN</ControlId>-->
      <!--<ControlId>btnInstancesHISTORY</ControlId>-->
      <!--<ControlId>btnInstancesDELEGATE</ControlId>-->
      <!--<ControlId>btnInstancesCLEARERROR</ControlId>-->
      <!--<ControlId>btnInstancesCLAIM</ControlId>-->
      <!--<ControlId>btnInstancesUNCLAIM</ControlId>-->
      <!--<ControlId>btnInstancesDELETE</ControlId>-->

  <!-- Tab VIEW -->
  <!--<ControlId>theRibbonTabVIEW</ControlId>-->
    <!--<ControlId>theRibbonGroupVIEW</ControlId>-->
      <!--<ControlId>cbxNavigation</ControlId>-->
      <!--<ControlId>cbxAdditionalInfo</ControlId>-->
    <!--<ControlId>theRibbonGroupVIEWERCLOSE</ControlId>-->
      <!--<ControlId>btnViewCLOSE</ControlId>-->

</HideList>

```

Tab

Customize an existing Tab.

Mandatory Attributes of Tab:

id	Specifies the Id of the Tab which should be customized.
----	---

Mandatory Elements of Tab:

Group	Specifies the Id of the new Group.
-------	------------------------------------

Possible IDs:

- theRibbonTabHOME
- theRibbonTabWORKFLOW
- theRibbonTabVIEW
- theRibbonTabTASKS

- theRibbonTabREPORTS

Example:

```
<Tab id="theRibbonTabHOME">
  <Group id="Therefore.NewGroup">
    ...
  </Group>
</Tab>
```

CustomTab

Add a new custom Tab.

Mandatory Attributes of CustomTab:

id	Specifies the Id of the Tab that should be customized.
----	--

Mandatory Elements of CustomTab:

Label	Specify a text for the custom Tab.
Group	Specifies the Id of the new Group.

Example:

```
<CustomTab id="Therefore.CustomTab">
  <Label resid="Therefore.CustomTab.Label"/>
  <Group id="Therefore.NewGroup">
    ...
  </Group>
</CustomTab>
```

Label

Specify a text for the custom Tab.

Mandatory Attributes of Label:

resid	Specifies a reference to a short string resource defined under the Resources element of the manifest.
-------	---

Group

Add a new Group.

Mandatory Attributes of Group:

id	Specifies the Id of the new Group.
----	------------------------------------

Mandatory Elements of Group:

Label	Specify a text for the new Group.
Control	Specifies the new Controls that should be added to this Group.

Example:

```

<Group id="Therefore.NewGroup">
  <Label resid="Therefore.NewGroup.Label"/>
  <Control xsi:type="the:Button" id="Therefore.SomeButton">
    ...
  </Control>
  <Control xsi:type="the:Button" id="Therefore.SomeOtherButton"
disabled="true">
    ...
  </Control>
</Group>

```

Control

Base type for "Button".

Mandatory Attributes of Control:

id	Specifies the id of the new Control.
disabled	Specifies if the Control should be disabled on initialization.

Mandatory Elements of Control:

Label	Specify a text for the new Control.
Tooltip	Specifies a tooltip for the new Control.

Example:

```

<Control xsi:type="the:Button" id="Therefore.FunctionButton">
  <Label resid="Therefore.FunctionButton.Label"/>
  <Tooltip resid="Therefore.FunctionButton.Description"/>
  <Icon>
    <bt:Image size="32" resid="Therefore.FunctionButton.IconUrl"/>
    <bt:ImageDisabled size="32"
resid="Therefore.FunctionButton.IconDisabledUrl"/>
  </Icon>
  <Action xsi:type="the:ExecuteFunction">
    <FunctionName>someFunction</FunctionName>
  </Action>
</Control>

```

Tooltip

Creates a tooltip for the new Control.

Mandatory Attributes of Tooltip:

resid	Specifies a reference to a long string resource defined under the Resources element of the manifest.
-------	--

Button

Adds a new Button

Mandatory Elements of Button:

Action	Specifies the type of Action in the Therefore client application.
Icon	Specifies the Icon for the Button.

Example:

```
<Control xsi:type="the:Button" id="Therefore.FunctionButton">
  <Label resid="Therefore.FunctionButton.Label"/>
  <Tooltip resid="Therefore.FunctionButton.Description"/>
  <Icon>
    <bt:Image size="32" resid="Therefore.FunctionButton.IconUrl"/>
    <bt:ImageDisabled size="32"
resid="Therefore.FunctionButton.IconDisabledUrl"/>
  </Icon>
  <Action xsi:type="the:ExecuteFunction">
    <FunctionName>someFunction</FunctionName>
  </Action>
</Control>
```

Icon

Specifies the Icon for the Button.

Mandatory Elements of Icon:

Image	Specifies the Image that should be displayed for the Button.
ImageDisabled	Same as Image, but for the disabled state.

Example:

```
<Icon>
  <bt:Image size="32"
resid="Therefore.FunctionButton.IconUrl"/>
```

```
<bt:ImageDisabled size="32"  
resid="Therefore.FunctionButton.IconDisabledUrl"/>  
</Icon>
```

Image and ImageDisabled

Mandatory Attributes of Image and ImageDisabled:

size	Specifies the size of the Image which should be displayed for the Button. There is only one size for now. Size = 32 has to be set.
resid	Specifies a reference to a Image resource defined under the Resources element of the manifest.

2.4.4.2.5 NavigatorRibbonBar

Derived from RibbonBarExtensionpoint.

By providing this Type, the Customization will be applied on the Main Page of Web Access.

2.4.4.2.6 ViewerRibbonBar

Derived from RibbonBarExtensionpoint.

By providing this Type, the Customization will be applied on the Viewer of Web Access.

Therefore customization should enable to retrieve current object information, like document, case, workflow instance and task numbers under Therefore™ WebViewer.

2.4.4.2.7 CaseViewerRibbonBar

Derived from RibbonBarExtensionpoint.

By providing this Type, the Customization will be applied on the CaseViewer of Web Access.

Therefore customization should enable to retrieve current object information, like document, case, workflow instance and task numbers under Therefore™ WebCaseManager.

2.4.4.2.8 TabStrip

Contains a collection of tabs.

Optional Elements of TabStrip:

EventTriggers	Specifies a collection of Events.
---------------	-----------------------------------

Example:

```
<ExtensionPoint xsi:type="the:TabStrip">
  <EventTriggers>
    ...
  </EventTriggers>
</ExtensionPoint>
```

2.4.4.2.9 Hitlist

Optional Elements of Hitlist:

EventTriggers	Specifies a collection of Events.
---------------	-----------------------------------

Example:

```
<ExtensionPoint xsi:type="the:Hitlist">
  <EventTriggers>
    ...
  </EventTriggers>
</ExtensionPoint>
```

2.4.5 EventTriggers

Specifies a collection of Events. Depending on the context, there will be different Event Types available to use.

Mandatory Elements of EventTriggers:

Event	Specifies the Event, which should be used.
-------	--

Example:

```
<EventTriggers>
  <Event xsi:type="the:OnInit">
    <Action xsi:type="the:ExecuteFunction">
      <FunctionName>someFunction</FunctionName>
    </Action>
  </Event>
</EventTriggers>
```

```

    </Action>
  </Event>
</EventTriggers>

```

2.4.5.1 Event

Specifies the Event, which should be used.

Mandatory Attributes of Event:

xsi:type	Specifies the type of the Event in the Therefore client application.
----------	--

Mandatory Elements of Event:

Action	Specifies the type of Action in the Therefore client application.
--------	---

Possible Types of Event:

Context	Event Types	Description
IndexDataDialog	OnInit	Remarks: Will not only occur OnInit but also on every Postback/Refresh.
	CategoryDialogButtonClick	Occurs when a button is clicked in the Toolbar and returns the Button name which was clicked.
	OnKeyPress	Occurs when a key is pressed.
	OnKeyDown	Occurs when a key state is changing from unpressed to pressed.
	CategoryFieldTableRowInserted	Remarks: for each row which is left after postback/refresh, this event will be triggered.
	CategoryFieldTableRowDeleted	Remarks: Will never be triggered. See "CategoryFieldTableRowInserted"
	CategoryFieldGotFocus	Occurs if a control got focus. Returns the Control Id/(Designer) Column Name.
	CategoryFieldLostFocus	Occurs if a control lost focus. Returns the Control Id/(Designer) Column Name.
	CategoryFieldChanged	Occurs if a controls content has changed. Returns the Control Id/(Designer) Column Name and the new value.

	CategoryFieldShowKeywordList	Occurs if a control shows available keywords (single/ multiple keyword field).
Hitlist	OnRowSelected	Occurs if a row is selected. Returns selected Document Numbers.
	OnRowDeselected	Occurs if a row is deselected. Returns selected Document Numbers.
	OnInitHitlist	Occurs if hitlist is initialized.
	OnPageLoad	Not implemented.
	OnGridCreated	Not implemented.
TabStrip	OnTabSelected	Occurs if a tab is selected/changed.

Example:

```
<EventTriggers>
  <Event xsi:type="the:OnInit">
    <Action xsi:type="the:ExecuteFunction">
      <FunctionName>someFunction</FunctionName>
    </Action>
  </Event>
  <Event xsi:type="the:OnKeyDown">
    <Action xsi:type="the:ExecuteFunction">
      <FunctionName>someOtherFunction</FunctionName>
    </Action>
  </Event>
</EventTriggers>
```

2.4.6 Action

Specifies the type of Action in the Therefore client application

Mandatory Attributes of Action:

xsi:type	Specifies the type of the Event in the Therefore client application. For now there is only one Type: ExecuteFunction
----------	--

Mandatory Elements for ExecuteFunction Type:

FunctionName	Specifies the function name which has to be called in the FunctionFile.
--------------	---

Example:

```
<Action xsi:type="the:ExecuteFunction">
  <FunctionName>someFunction</FunctionName>
</Action>
```

2.4.7 Resources

Defines a collection of controls to be added.

Mandatory Elements of Resources:

Images	Specifies Image Resources
Urls	Specifies Url Resources
ShortStrings	Specifies ShortString Resources. Up to 125 characters.
LongStrings	Specifies LongString Resources. Up to 250 characters.

Example:

```

<Resources>
  <bt:Images>
    ...
  </bt:Images>
  <bt:Urls>
    ...
  </bt:Urls>
  <bt:ShortStrings>
    ...
  </bt:ShortStrings>
  <bt:LongStrings>
    ...
  </bt:LongStrings>
</Resources>

```

2.4.7.1 Images

Specifies Image Resources

Mandatory Elements of Image:

Image	Specifies an Image to the Resources.
-------	--------------------------------------

2.4.7.1.1 Image

This Element is locale aware. Refer to “LocaleAwareSetting”.

Mandatory Attributes of Image:

id	Specifies an Id for the Image. This can be used in “resid” attribute.
----	---

2.4.7.2 URLs

Defines a collection of URLs.

Mandatory Elements of URLs:

URL	Specifies a URL to the Resources.
-----	-----------------------------------

2.4.7.2.1 URL

This Element is locale aware. Refer to "LocaleAwareSetting".

Mandatory Attributes of URL:

id	Specifies an Id for the Url. This can be used in "resid" attribute.
----	---

2.4.7.3 ShortStrings and LongStrings

With ShortStrings, each String contains a maximum of 125 characters and with LongStrings each String contains a maximum of 250 characters.

Mandatory Elements of ShortStrings and Longstrings:

String	Specifies a String to the Resources.
--------	--------------------------------------

2.4.7.3.1 String

This Element is locale aware. Refer to "LocaleAwareSetting".

Mandatory Attributes of String:

id	Specifies an Id for the String. This can be used in "resid" attribute.
----	--

2.4.8 LocaleAwareSetting

Mandatory Attributes of LocaleAwareSetting:

DefaultValue	Specifies the default value for this setting, expressed for the locale specified in the "DefaultLocale" element.
--------------	--

Optional Elements of LocaleAwareSetting:

Override	Provides a way to specify the value of this setting for an additional locale
----------	--

2.4.8.1 Override

Provides a way to specify the value of a setting for an additional locale.

Mandatory Attributes of Override:

Locale	Specifies the culture name of the locale for this override
DefaultValue	Specifies value of the setting expressed for the specified locale

2.4.9 FunctionFile

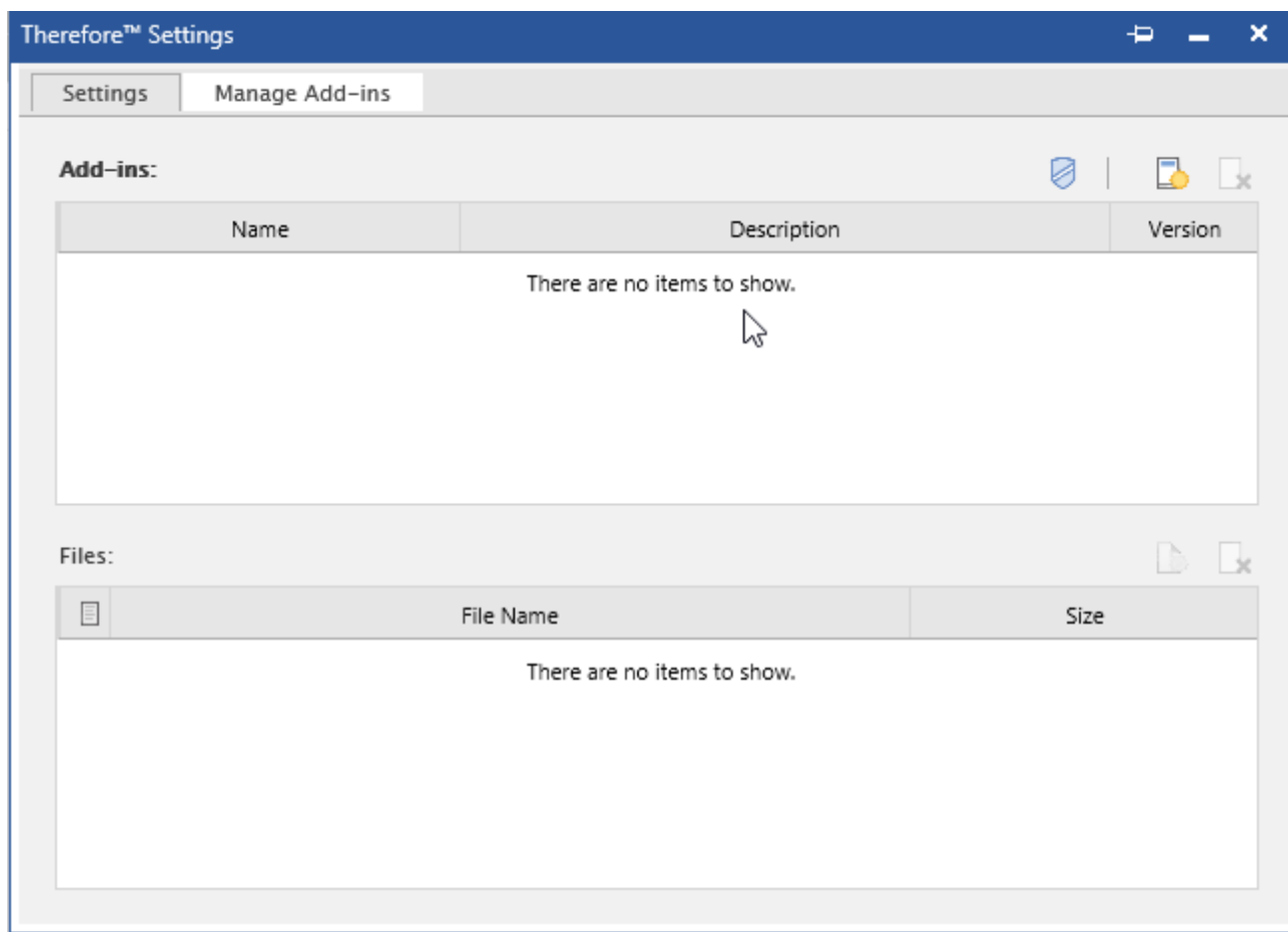
Then you have to generate your own function file, which must use the Therefore framework:

```
<!DOCTYPE html>
<html>
<head>
  <title>FunctionFile</title>
  <script src="//$SERVER/TWA/AddIn/therefore.js"></script>
  <script>
    Custom code...
  </script>
</head>
<body>
  body of function files won't be displayed
</body>
</html>
```

Between the <script> tags the developer can write their own "Custom code" for the customization. Each and every function must be defined in order to be used in the manifest file.

2.4.10 Upload and Managing add-ins

Once the manifest and function files (and images if buttons are added to the RibbonBar) have been obtained, the files can be uploaded via the WebClient. Specific users can be granted permission to "Manage Add-ins" so that the developer who creates the customization can upload and test the product through live coding (on the fly).



If the “Manage Add-ins” tab is available under the settings dialog the add-ins can be accessed. At the top-right hand corner of the dialog, there are three icons:

- Security: Configure and manage add-ins permissions
- Add Add-in: Upload add-in files
- Delete Add-in: Remove add-in files

The top grid lists out the currently uploaded add-ins (XML)

The bottom grid holds all the files, which belong to the currently selected add-in (HTML, JPEG, etc.)

On the right-hand side above the bottom grid there are two more icons. These buttons enable files to be added to the currently selected add-in.

2.4.11 Demo and Testing

We are publishing some templates to demonstrate what can be done with customization e.g:

- Template: Manipulate an existing button on RibbonBar
- Template: Add new button to RibbonBar

- Template: Calculated fields on category dialog

The templates can be used for testing – but must be updated to have a valid IDs, names etc. This can be implemented by reading and following the instructions of the read.me file. Once you have updated the template you can upload and activate on your system and can test the customization.

2.4.11.1 Template 1: Manipulate an existing button on RibbonBar

```
<?xml version="1.0" encoding="utf-8" ?>
<the:ThereforeAddIn
  xmlns:the="http://schemas.therefore.net/thereforeaddinmanifest/0.1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:bt="http://schemas.therefore.net/thereforebasictypes/0.1">
  <Id>73577357-abcd-7357-ffff-735773570003</Id>
  <Version>0.1</Version>
  <ProviderName>Therefore</ProviderName>
  <DefaultLocale>en-US</DefaultLocale>
  <DisplayName DefaultValue="Test RibbonBar-ManipulateButtons" />
  <Description DefaultValue="The Description" />
  <IconUrl DefaultValue="/TWA/Client/Images/therefore_32.png" />
  <Hosts>
    <Host xsi:type="the:Navigator">
      <DesktopFormFactor>
        <FunctionFile resid="Therefore.FunctionFile.Url"/>
        <ExtensionPoint xsi:type="the:NavigatorRibbonBar">
          <HideList>
            <!-- RibbonBar ApplicationMenu (Context Menu) -->
            <!--<ControlId>appMenuItemNEW</ControlId>-->
            <!--<ControlId>appMenuItemNEWDOC</ControlId>-->
            <!--<ControlId>appMenuItemNEWCASE</ControlId>-->
            <!--<ControlId>appMenuItemCOPY</ControlId>-->
            <!--<ControlId>appMenuItemPRINT</ControlId>-->
            <ControlId>appMenuItemRECYCLEBIN</ControlId>
            <!--<ControlId>appMenuItemREFRESHTREEVIEW</ControlId>-->
            <!--<ControlId>appMenuItemADMINISTRATIVETOOLS</ControlId>-->
            <!--<ControlId>appMenuItemOUTOFFOFFICE</ControlId>-->
            <!--<ControlId>appMenuItemDOWNLOADGADGET</ControlId>-->

            <!-- RibbonBar -->

            <!-- Tab HOME -->
            <!--<ControlId>theRibbonTabHOME</ControlId>-->
            <!--<ControlId>theRibbonGroupSEARCH</ControlId>-->
            <!--<ControlId>btnSEARCH</ControlId>-->
            <!--<ControlId>btnSearchNEWSEARCH</ControlId>-->
            <!--<ControlId>btnSearchNEWCROSSCATEGORYSEARCH</ControlId>-->
            <!--<ControlId>btnOpenSearch</ControlId>-->
            <!--<ControlId>theRibbonGroupFULLTEXT</ControlId>-->
            <!--<ControlId>txtbxQuickSearch</ControlId>-->
            <!--<ControlId>cbxDocumentContent</ControlId>-->
            <!--<ControlId>cbxIndexData</ControlId>-->
            <!--<ControlId>btnFullTextADVANCEDSEARCH</ControlId>-->
```

```

<!--<ControlId>theRibbonGroupHITLIST</ControlId>-->
  <!--<ControlId>btnHitlistMODE</ControlId>-->
  <!--<ControlId>btnHitlistREFRESH</ControlId>-->
  <!--<ControlId>btnHitlistPRINT</ControlId>-->
  <!--<ControlId>btnHitlistCOPY</ControlId>-->
<!--<ControlId>theRibbonGroupDOCUMENTS</ControlId>-->
  <!--<ControlId>btnDocumentsTEMPLATES</ControlId>-->
  <!--<ControlId>btnDocumentsVIEW</ControlId>-->
  <!--<ControlId>btnDocumentsPRINT</ControlId>-->
  <!--<ControlId>btnDocumentsINDEXDATA</ControlId>-->
  <!--<ControlId>btnDocumentsEXPORT</ControlId>-->
    <!--<ControlId>btnDocumentsExportEXPORT</ControlId>-->
    <!--<ControlId>btnDocumentsExportSEND</ControlId>-->
  <!--<ControlId>btnDocumentsLINKS</ControlId>-->
  <!--<ControlId>btnDocumentsSECURITY</ControlId>-->
  <!--<ControlId>btnDocumentsHISTORY</ControlId>-->
  <!--<ControlId>btnDocumentsDELETE</ControlId>-->
<!--<ControlId>theRibbonGroupPROCESS</ControlId>-->
  <!--<ControlId>btnProcessNEWTASK</ControlId>-->
  <!--<ControlId>btnProcessSTART</ControlId>-->
  <!--<ControlId>btnProcessLINKTO</ControlId>-->

<!-- Tab WORKFLOW -->
<!--<ControlId>theRibbonTabWORKFLOW</ControlId>-->
  <!--<ControlId>theRibbonGroupVIEWS</ControlId>-->
    <!--<ControlId>btnViewsMYINSTANCES</ControlId>-->
    <!--<ControlId>btnViewsALLINSTANCES</ControlId>-->
    <!--<ControlId>btnViewsCONFIGURE</ControlId>-->
  <!--<ControlId>theRibbonGroupINSTANCES</ControlId>-->
    <!--<ControlId>btnInstancesOPEN</ControlId>-->
    <!--<ControlId>btnInstancesHISTORY</ControlId>-->
    <!--<ControlId>btnInstancesDELEGATE</ControlId>-->
    <!--<ControlId>btnInstancesCLEARERROR</ControlId>-->
    <!--<ControlId>btnInstancesCLAIM</ControlId>-->
    <!--<ControlId>btnInstancesUNCLAIM</ControlId>-->
    <!--<ControlId>btnInstancesDELETE</ControlId>-->

<!-- Tab VIEW -->
<!--<ControlId>theRibbonTabVIEW</ControlId>-->
  <!--<ControlId>theRibbonGroupVIEW</ControlId>-->
    <!--<ControlId>cbxNavigation</ControlId>-->
    <!--<ControlId>cbxAdditionalInfo</ControlId>-->
  <!--<ControlId>theRibbonGroupVIEWERCLOSE</ControlId>-->
    <!--<ControlId>btnViewCLOSE</ControlId>-->

</HideList>
<Tab id="theRibbonTabHOME">
  <Group id="Therefore.GroupManButton">
    <Label resid="Therefore.GroupManButton.Label"/>
    <Control xsi:type="the:Button" id="Therefore.ButtonToggle"
disabled="false">
      <Label resid="Therefore.ButtonToggle.Label"/>
      <Tooltip resid="Therefore.ButtonToggle.Tooltip"/>
      <Icon>

```

```

        <bt:Image size="32" resid="Therefore.Button.Image"/>
        <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
        </Icon>
        <Action xsi:type="the:ExecuteFunction">
            <FunctionName>buttonToggleClick</FunctionName>
        </Action>
    </Control>
    <Control xsi:type="the:Button" id="Therefore.ButtonToToggle"
disabled="true">
        <Label resid="Therefore.ButtonToToggle.Label"/>
        <Tooltip resid="Therefore.ButtonToToggle.Tooltip"/>
        <Icon>
            <bt:Image size="32" resid="Therefore.Button.Image"/>
            <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
        </Icon>
        <Action xsi:type="the:ExecuteFunction">
            <FunctionName>buttonClickMe</FunctionName>
        </Action>
    </Control>
</Group>
</Tab>
</ExtensionPoint>
</DesktopFormFactor>
</Host>
</Hosts>
<Resources>
    <bt:Images>
        <bt:Image id="Therefore.Button.Image"
DefaultValue="/TWA/Client/Images/ribbon_workflow_instance_process_32.png" />
        <bt:Image id="Therefore.Button.ImageDisabled"
DefaultValue="/TWA/Client/Images/ribbon_disabled_workflow_instance_process_32.png"
/>
    </bt:Images>
    <bt:Urls>
        <bt:Url id="Therefore.FunctionFile.Url"
DefaultValue="/TWA/Client/AddIn/TWA/73577357-abcd-7357-ffff-
735773570003/FunctionFile.html" />
    </bt:Urls>
    <bt:ShortStrings>
        <bt:String id="Therefore.GroupManButton.Label" DefaultValue="Manipulate
Buttons" />
        <bt:String id="Therefore.ButtonToggle.Label" DefaultValue="Toggle" />
        <bt:String id="Therefore.ButtonToToggle.Label" DefaultValue="Click Me" />
    </bt:ShortStrings>
    <bt:LongStrings>
        <bt:String id="Therefore.ButtonToggle.Tooltip" DefaultValue="Toggle Tooltip"
/>
        <bt:String id="Therefore.ButtonToToggle.Tooltip" DefaultValue="Click Me
Tooltip" />
    </bt:LongStrings>
</Resources>
</the:ThereforeAddIn>

```

2.4.11.2 Template 2: Add new button to RibbonBar

```
<?xml version="1.0" encoding="utf-8" ?>
<the:ThereforeAddIn
  xmlns:the="http://schemas.therefore.net/thereforeaddinmanifest/0.1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:bt="http://schemas.therefore.net/thereforebasictypes/0.1">
  <Id>73577357-abcd-7357-ffff-735773570002</Id>
  <Version>0.1</Version>
  <ProviderName>Therefore</ProviderName>
  <DefaultLocale>en-US</DefaultLocale>
  <DisplayName DefaultValue="Test RibbonBar-AddButtons" />
  <Description DefaultValue="The Description" />
  <IconUrl DefaultValue="/TWA/Client/Images/therefore_32.png" />
  <Hosts>
    <Host xsi:type="the:Navigator">
      <DesktopFormFactor>
        <FunctionFile resid="Therefore.FunctionFile.Url"/>
        <ExtensionPoint xsi:type="the:NavigatorRibbonBar">
          <Tab id="theRibbonTabHOME">
            <Group id="Therefore.NewGroup1">
              <Label resid="Therefore.NewGroup1.Label"/>
              <Control xsi:type="the:Button" id="Therefore.Button1">
                <Label resid="Therefore.Button1.Label"/>
                <Tooltip resid="Therefore.Button1.Tooltip"/>
                <Icon>
                  <bt:Image size="32" resid="Therefore.Button.Image"/>
                  <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
                </Icon>
                <Action xsi:type="the:ExecuteFunction">
                  <FunctionName>button1Click</FunctionName>
                </Action>
              </Control>
              <Control xsi:type="the:Button" id="Therefore.Button2">
                <Label resid="Therefore.Button2.Label"/>
                <Tooltip resid="Therefore.Button2.Tooltip"/>
                <Icon>
                  <bt:Image size="32" resid="Therefore.Button.Image"/>
                  <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
                </Icon>
                <Action xsi:type="the:ExecuteFunction">
                  <FunctionName>button2Click</FunctionName>
                </Action>
              </Control>
            </Group>
          </Tab>
          <CustomTab id="Therefore.CustomTab1">
            <Label resid="Therefore.CustomTab1.Label"/>
            <Group id="Therefore.NewGroup2">
```

```

        <Label resid="Therefore.NewGroup2.Label"/>
        <Control xsi:type="the:Button" id="Therefore.Button3">
            <Label resid="Therefore.Button3.Label"/>
            <Tooltip resid="Therefore.Button3.Tooltip"/>
            <Icon>
                <bt:Image size="32" resid="Therefore.Button.Image"/>
                <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
            </Icon>
            <Action xsi:type="the:ExecuteFunction">
                <FunctionName>button3Click</FunctionName>
            </Action>
        </Control>
    </Group>
</CustomTab>
</ExtensionPoint>
<ExtensionPoint xsi:type="the:ViewerRibbonBar">
    <Tab id="theRibbonTabVIEW">
        <Group id="Therefore.NewGroup3">
            <Label resid="Therefore.NewGroup3.Label"/>
            <Control xsi:type="the:Button" id="Therefore.Button4">
                <Label resid="Therefore.Button4.Label"/>
                <Tooltip resid="Therefore.Button4.Tooltip"/>
                <Icon>
                    <bt:Image size="32" resid="Therefore.Button.Image"/>
                    <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
                </Icon>
                <Action xsi:type="the:ExecuteFunction">
                    <FunctionName>button4Click</FunctionName>
                </Action>
            </Control>
        </Group>
    </Tab>
    <CustomTab id="Therefore.CustomTab2">
        <Label resid="Therefore.CustomTab2.Label"/>
        <Group id="Therefore.NewGroup4">
            <Label resid="Therefore.NewGroup4.Label"/>
            <Control xsi:type="the:Button" id="Therefore.Button5">
                <Label resid="Therefore.Button5.Label"/>
                <Tooltip resid="Therefore.Button5.Tooltip"/>
                <Icon>
                    <bt:Image size="32" resid="Therefore.Button.Image"/>
                    <bt:ImageDisabled size="32"
resid="Therefore.Button.ImageDisabled"/>
                </Icon>
                <Action xsi:type="the:ExecuteFunction">
                    <FunctionName>button5Click</FunctionName>
                </Action>
            </Control>
        </Group>
    </CustomTab>
</ExtensionPoint>
</DesktopFormFactor>

```

```

    </Host>
  </Hosts>
  <Resources>
    <bt:Images>
      <bt:Image id="Therefore.Button.Image"
DefaultValue="/TWA/Client/Images/ribbon_workflow_instance_process_32.png" />
      <bt:Image id="Therefore.Button.ImageDisabled"
DefaultValue="/TWA/Client/Images/ribbon_disabled_workflow_instance_process_32.png"
/>
    </bt:Images>
    <bt:Urls>
      <bt:Url id="Therefore.FunctionFile.Url"
DefaultValue="/TWA/Client/AddIn/TWA/73577357-abcd-7357-ffff-
735773570002/FunctionFile.html" />
    </bt:Urls>
    <bt:ShortStrings>
      <bt:String id="Therefore.CustomTab1.Label" DefaultValue="CustomTab1" />
      <bt:String id="Therefore.CustomTab2.Label" DefaultValue="CustomTab2" />
      <bt:String id="Therefore.NewGroup1.Label" DefaultValue="NewGroup1" />
      <bt:String id="Therefore.NewGroup2.Label" DefaultValue="NewGroup2" />
      <bt:String id="Therefore.NewGroup3.Label" DefaultValue="NewGroup3" />
      <bt:String id="Therefore.NewGroup4.Label" DefaultValue="NewGroup4" />
      <bt:String id="Therefore.Button1.Label" DefaultValue="Button1" />
      <bt:String id="Therefore.Button2.Label" DefaultValue="Button2" />
      <bt:String id="Therefore.Button3.Label" DefaultValue="Button3" />
      <bt:String id="Therefore.Button4.Label" DefaultValue="Button4" />
      <bt:String id="Therefore.Button5.Label" DefaultValue="Button5" />
    </bt:ShortStrings>
    <bt:LongStrings>
      <bt:String id="Therefore.Button1.Tooltip" DefaultValue="Button1 Tooltip" />
      <bt:String id="Therefore.Button2.Tooltip" DefaultValue="Button2 Tooltip" />
      <bt:String id="Therefore.Button3.Tooltip" DefaultValue="Button3 Tooltip" />
      <bt:String id="Therefore.Button4.Tooltip" DefaultValue="Button4 Tooltip" />
      <bt:String id="Therefore.Button5.Tooltip" DefaultValue="Button5 Tooltip" />
    </bt:LongStrings>
  </Resources>
</the:ThereforeAddIn>

```

2.4.11.3 Template 3: Calculated fields on category dialog

```

<?xml version="1.0" encoding="utf-8" ?>
<the:ThereforeAddIn
  xmlns:the="http://schemas.therefore.net/thereforeaddinmanifest/0.1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:bt="http://schemas.therefore.net/thereforebasictypes/0.1">
  <Id>73577357-abcd-7357-ffff-735773570001</Id>
  <Version>0.1</Version>
  <ProviderName>Therefore</ProviderName>
  <DefaultLocale>en-US</DefaultLocale>
  <DisplayName DefaultValue="Test IndexDataDialog-CalcSum" />
  <Description DefaultValue="The Description" />
  <IconUrl DefaultValue="/TWA/Client/Images/therefore_32.png" />

```



```
<Hosts>
  <Host xsi:type="the:Navigator">
    <DesktopFormFactor>
      <FunctionFile resid="Therefore.FunctionFile.Url"/>
      <ExtensionPoint xsi:type="the:CategoryIndexDataDialog"
CategoryNo="[CategoryNumber]">
        <EventTriggers>
          <Event xsi:type="the:OnKeyDown">
            <Action xsi:type="the:ExecuteFunction">
              <FunctionName>calcFunctionTableSumVal</FunctionName>
            </Action>
          </Event>
          <Event xsi:type="the:OnInit">
            <Action xsi:type="the:ExecuteFunction">
              <FunctionName>calcFunctionTableSumVal</FunctionName>
            </Action>
          </Event>
        </EventTriggers>
      </ExtensionPoint>
    </DesktopFormFactor>
  </Host>
</Hosts>
<Resources>
  <bt:Urls>
    <bt:Url id="Therefore.FunctionFile.Url"
DefaultValue="[UrlFunctionFile]" />
  </bt:Urls>
</Resources>
</the:ThereforeAddIn>
```

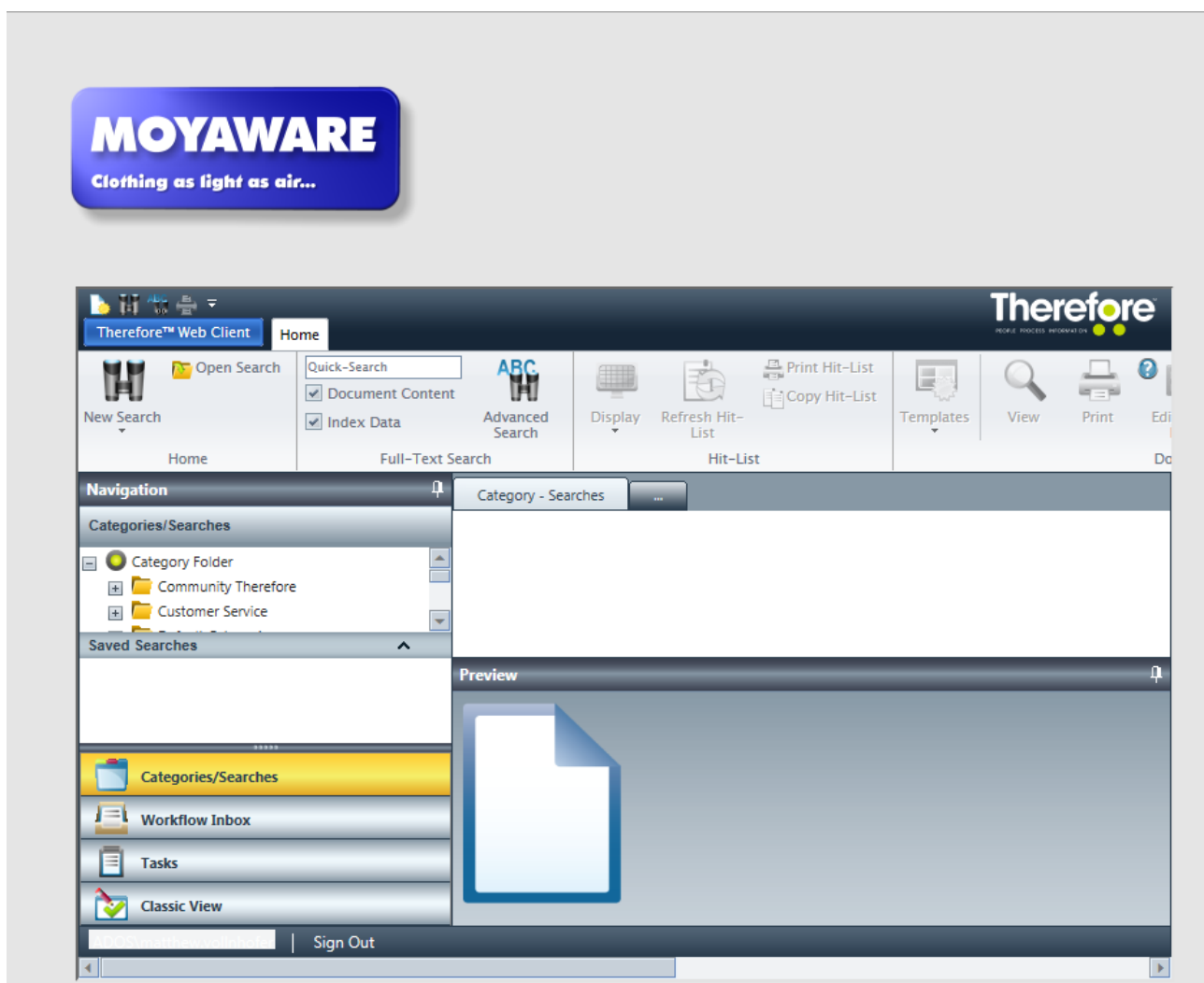
3. Customizing Aesthetics

3.1 Embedding into another website

It is possible to embed Therefore™ into another website in two ways.

Fully embedded

Here the full Therefore™ Web Access is simply displayed within another page. Therefore™ Web Access behaves in exactly the same way as when in its own browser window. To embed simply use the standard Therefore™ Web Access URL (http://server_name/TWA). It is also possible to [replace the Therefore logo](#) or [customizing the skin](#).



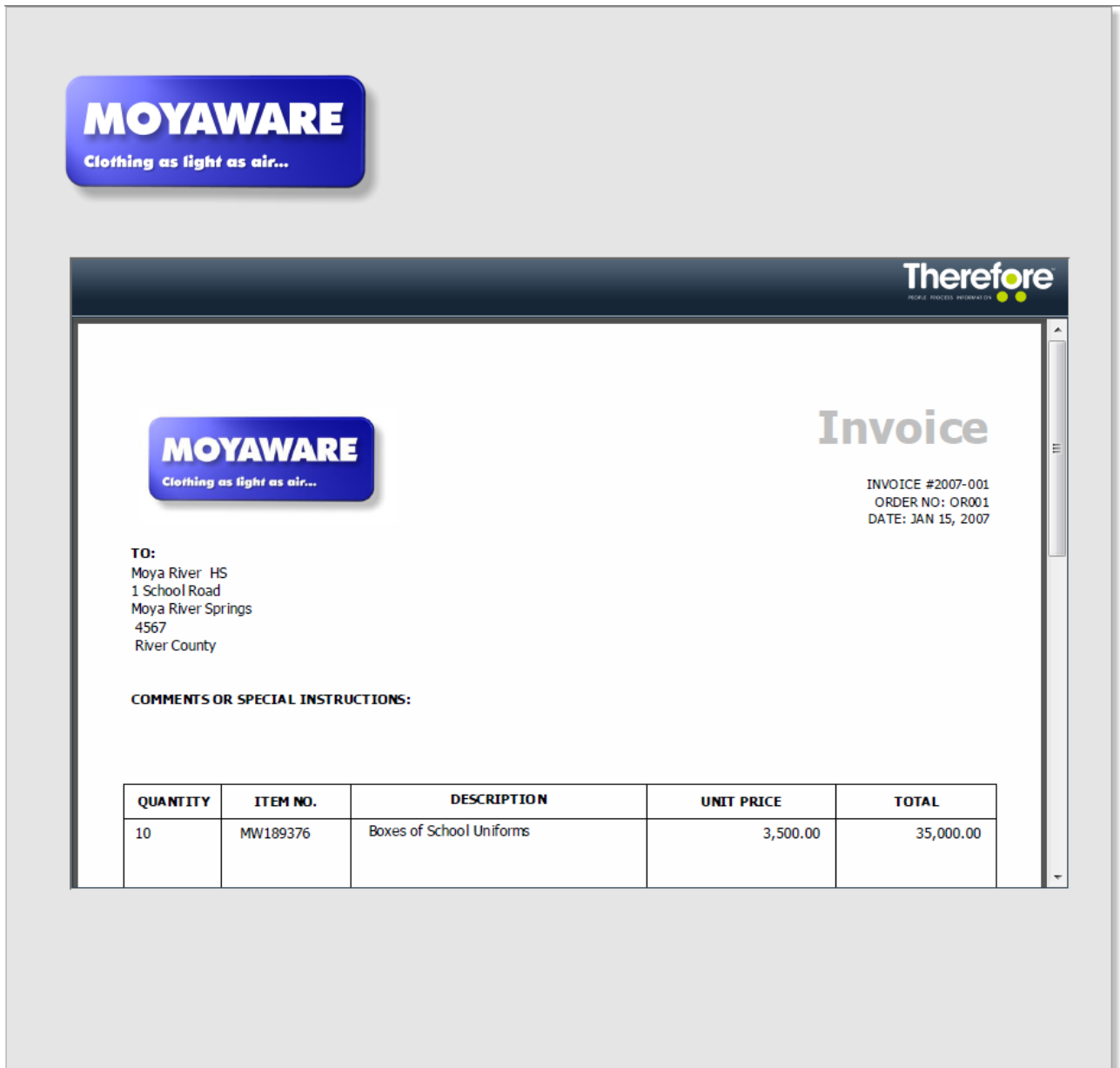
When a document is opened, it will be opened in a new browser instance, but the communication between the Therefore™ Web Viewer and Therefore™ Web Navigator still exists.

Embed a certain document for viewing

Where only a specific document should be embedded for viewing. For example to give a customer a view of their invoice then the following URL can be used:

Therefore™ On-Premise: http://server_name/TWA/Viewer.aspx?DocNo=xx&embed=true

Therefore™ Online: <https://tenantname.thereforeonline.com/Viewer.aspx?DocNo=X>



- All files in the document will be converted to a single PDF for viewing. Hence a PDF viewer is required.
- The document can also be specified using the workflow instance no. or the task no.
<http://server/TWA/Viewer.aspx?InstanceNo=xx&embed=true>
<http://server/TWA/Viewer.aspx?TaskNo=xx&embed=true>
- It is also possible to [replace the Therefore logo](#).



3.2 Changing the logo

You can specify your own logo, by replacing the existing logo files in the following locations:

WebNavigator & WebViewer on PCs and Tablets:

UI logo: "
\\SERVER\\inetpub\\wwwroot\\TWA\\Client\\Styles\\SKINS\\Therefore\\Common\\logoCompanyWhite.png"

Suggested size is: 130x38

Login page logo: "
\\SERVER\\inetpub\\wwwroot\\TWA\\Client\\Styles\\SKINS\\Therefore\\LoginPage\\imgBrandingLogo.png"

Suggested size is: 160x160



- File name and file type cannot be changed.
- There is no size limit, but larger images will take longer to load.



Multi-tenant systems

In the SKINS folder, create a new folder and give it the same name as the tenant. Then copy the contents of the Therefore folder into this new folder.

3.3 Creating your own skin

This is done by editing or replacing the existing skin: "~\\Styles\\SKINS\\Therefore".



- CSS knowledge is required and the required graphics need to be created.
- We strongly recommend that you keep a backup of the original skin.

For example to have red font color on the search dialog the cbDialogFont style-sheet rule must be changed in the "CategoryDialog.css" file:

```
.cbDialogFont
{
  color: red !important;
}
```



Multi-tenant systems

In the SKINS folder, create a new folder and give it the same name as the tenant. Then copy the contents of the Therefore folder into this new folder.

4. Other

4.1 Download link

A download link in the Therefore™ Options dialog can be created as follows:

1. Create a folder called "Download" in the Therefore™ Web Access root folder (by default this the TWA folder).
2. Copy the files for downloading to this directory.
3. Create an xml file called "download.xml" in the following format:

```
<?xml version="1.0" encoding="utf-8" ?>
<downloads>
  <download>
    <file>Therefore 2012 Client 32-Bit.exe</file>
    <link>Download - Therefore 2012 Client 32-Bit</link>
    <file>File 2..</file>
    <link>Download File2..</link>
  </download>
</downloads>
```

4.2 Information Link

The information link can be activated as follows:

1. Set the EnableInfoButton key value to true in the web application's web.config file:

```
<add key="EnableInfoButton" value="true" />
```

4.3 Compact URL for Multi-tenant systems

The default Therefore™ WebAccess URL looks like this:

http://server_name/Tenant Web Application Name/...

However it is possible to compact this URL using subdomain mode. Then the URL will look like:

http://Tenant Web Application name.server_name/...

To change to subdomain mode, set the SubdomainMode key value to true in the Client web application's "web.config" file:

```
<add key="SubdomainMode" value="true" />
```

4.4 Using multiple MFP Manager instances in parallel with UniFlow Login Manager

The encryption key for secure password exchange between the Therefore MFP Manager and UniFlow is currently stored locally for each machine.

This would cause a problem in a high availability environment with two or more DMZ servers, because each DMZ server would have its own public/private key pair.

As a solution to this problem, it is possible to configure a Microsoft Azure KeyVault as secure way to share encryption keys among different instances of the MFP Manager.

The following settings need to be configured to use a KeyVault for key storage

Additional Keys in the MFP Manager config file:

```
<!-- Azure Key Vault configuration (activate the commented out lines and configure
valid parameters) -->
<!--<add key="Azure.AppId" value="" />
<add key="Azure.AuthKey" value="" />
<add key="Azure.KeyVaultUri" value="" />-->
<!--Optional timeout in milliseconds for Azure requests: <add
key="Azure.RequestTimeout" value="" />-->
```

Key permissions in the MS Azure Portal

- The application (configured by Azure.AppId) requires the certain access policies in the KeyVault (under 'Key Permissions').
 - o Key Management Operations: Get, Create
 - o Cryptographic Operations: Decrypt

Additional information for Azure configuration

- The name for the encryption key is hardcoded to "TheMFPManager".
- Although the key gets stored in Azure, the public part gets still written to the "PublicKey.txt" and can be used to configure UniFlow.

4.5 Using multiple Web Service & IIS instances in parallel with Therefore Web Forms

The user name and access token for anonymous web form users is currently stored in the registry of the WebService instance.

This would cause a problem in a high availability environment with two or more DMZ servers, because each DMZ server would have its own user name and access token.

As a solution to this problem, it is possible to configure a Microsoft Azure KeyVault as secure way to share this information among different instances of the WebService.

The following settings need to be configured to use a KeyVault for storing the form user credentials

```
Additional Keys in the web.config file:
<!-- Azure Key Vault configuration (activate the commented out lines and configure
valid parameters) -->
<!--<add key="Azure.AppId" value="" />
<add key="Azure.AuthKey" value="" />
<add key="Azure.KeyVaultUri" value="" />-->
<!--Optional timeout in milliseconds for Azure requests: <add
key="Azure.RequestTimeout" value="" />-->
```

Secret permissions in the MS Azure Portal

Secret Management Operations: Get and Set.

4.6 Sign-on

Therefore™ Web Access supports single sign-on for Windows® users using Internet Explorer® and Mozilla Firefox, but this needs to be supported by the browser too. Otherwise a customizable login page is displayed before the application can be accessed. The user is asked to enter their user name and password (see table below). Furthermore, for access to be granted, a user license is required.

However, it is also possible to configure auto-login with a pre-defined default user account. In the login screen only one button, Anonymous Login will be available. This can be configured in the file "Web.config" by replacing <appSettings/> with the following:

```
<appSettings>
  <add key="AutoLoginMode" value="Interactive"></add>
  <add key="AutoLoginUser" value="MOYAWARE\jackson.a"></add>
  <add key="AutoLoginPwd" value="123456"></add>
</appSettings>
```

If no login screen should be shown:

```
<appSettings>
  <add key="AutoLoginMode" value="Automatic"></add>
  <add key="AutoLoginUser" value="MOYAWARE\jackson.a"></add>
  <add key="AutoLoginPwd" value="123456"></add>
</appSettings>
```

4.6.1 Automatically log in

On the Web Access login page, users can choose to stay signed in in order to avoid logging in manually to each session. An administrator can configure certain settings around this feature by editing the Client "web.config" file under \\SERVER\inetpub\wwwroot\TWA\Client.

```
<appSettings>
  <add key="theTimeOut" value="20" /><!-- SESSION EXPIRE TIME IN MINUTES -
  DEFAULT IS 20 MINS -->
  <add key="theStaySignedInEnabled" value="true" /><!-- ENABLE/DISABLE "STAY
  SIGNED IN" FEATURE -->
  <add key="theStaySignedInTimeOut" value="10" /><!-- COOKIE EXPIRE TIME IN DAYS
  FOR "STAY SIGNED IN" FEATURE - DEFAULT IS 10 DAYS -->
</appSettings>
```

theTimeOut:

Holds the expiry time for sessions. If there is no client-side activity for amount of time configured, the user will be redirected to the Web Access login page. The value is stored in minutes.

theStaySignedInEnabled:

Determines whether or not the checkbox "Automatically log in" is available to users. The default value is true. When set to false, the checkbox is disabled.

theStaySignedInTimeOut:

Holds the expiry time for the cookies if a user has chosen to automatically log in. This determines how long the credentials can remain saved for automatic login. The value is stored in days. After the cookies expire users must enter their credentials once more when logging in.

4.7 Disable Password Change

For users created in Therefore™ it is possible to disable the user's ability to change their password as follows:

Under the DisablePasswordChange key value, in the web application's web.config file, fill in the usernames and or email addresses of the users who should not be allowed to change their passwords:

```
<add key="DisablePasswordChange" value="jackson.a@moyaware.com;smithson.j@moyaware.com;msmith" />
```

4.8 Configure DNS Server Access

DNS Server calls be configured via the web.config file located under the TWA folder: [\SERVER\inetpub\wwwroot\TWA\web.config](#)

```
<add key="EnableGetDNSHostEntry" value="true" /><!-- ENABLE TO ACCESS DNS SERVER FOR THE IP ADDRESS WHICH IS ASSOCIATED WITH A HOST NAME OR IP - DEFAULT VAULE: "true" -->
```

The value is set to True by default. Access to the DNS Server will be denied if the value is set to False. In this case, the IP address will be used, instead.

Denying DNS Access can help improve Therefore™ Web Client performance if the DNS Server is misconfigured or very slow. Otherwise this is not recommended, as this method causes a distinct disadvantage: if you are using both installed and web-based Therefore™ applications (such as Therefore™ Navigator and Therefore™ Web Access), two licenses will be consumed, one for the client using the host name (installed) and one for the client using IP (web).



As the web.config file is located directly under the TWA level, it is a global setting will be used not only by WebClient, but also by WebMeap (MFP) and Portal.

4.9 Enable Windows Authentication on Firefox

Windows authentication can be enabled on Firefox to allow single-on. Follow these steps to enabled this:

1. Open Mozilla Firefox.
2. Type about:config into the address bar.
3. Type network.automatic into the filter box.
4. Modify network.automatic-ntlm-auth.trusted-uris by double-clicking on the row and enter: <http://servername>, <http://servername/TWA>, <http://servername/TWA/Client>. For multi-tenant systems, change TWA to the name of the tenant.
5. Set network.automatic-ntlm-auth.allow-non-fqdn to TRUE.

Firefox will now use windows authentication for the configured website, so no login page will be displayed when accessing Therefore™ Web Access.

4.10 URL Actions

There are a number of actions that can be performed using URLs. These could, for example, be triggered from a 3rd party application that has the needed integration capability.



Executing a search

Therefore™ On-Premise:

http://server/TWA/TheGetDoc.aspx?CtgryNo=X&FieldName1=filter_value&FieldName2=filter_value

Therefore™ Online:

https://tenantname.thereforeonline.com/TheGetDoc.aspx?CtgryNo=X&FieldName1=filter_value&FieldName2=filter_value

CtgryNo:

Replace X with the correct category ID.

FieldName1:

Replace this value with the input field's column name. The column name can be found under the category definition.

Replace the filter value with natural text. Spaces and operands can be used.

When filtering multiple keyword fields, use the following syntax:

TWF+FieldNo=value1, value2

The FieldNo can be found in the database in the TheCtgryFields table. For example:

&TWF123=january, february

To add more fields, adapt the URL by adding &FieldName=filter value after the last value (see example for FieldName2 above)

If the search returns only one item, it is automatically opened in the Web Viewer. Otherwise the hit-list will be displayed in the Web Navigator.



Executing a full-text search

Therefore™ On-Premise:

<http://server/TWA/theGetDoc.aspx?fulltext=true&fulltextfilter=string>

Therefore™ Online:

<https://tenantname.thereforeonline.com/theGetDoc.aspx?fulltext=true&fulltextfilter=string>

Replace the string with the actual search string.



Executing an index data field search

Therefore™ On-Premise:

<http://server/TWA/TheGetDoc.aspx?CtgryNo=X&Index1=Value1&Index2=Value2>

Therefore™ Online:

<https://tenantname.thereforeonline.com/TheGetDoc.aspx?CtgryNo=X&Index1=Value1&Index2=Value2>

Replace X with the category number, Index1, Index2 etc. with the actual column name of the index field and Value1, Value2 etc. with the search values. If only one document is found then it will be opened in the Viewer.



Open a specific document in the Therefore™ Viewer

Therefore™ On-Premise:

<http://server/TWA/Viewer.aspx?DocNo=X>

Therefore™ Online:

<https://tenantname.thereforeonline.com/Viewer.aspx?DocNo=X>

Replace X with the document number and version number.

If you want to return a specific document version specify X.Y. Y is the version number.



Opening a specific Therefore™ Task

Therefore™ On-Premise:

<http://server/TWA/Viewer.aspx?TaskNo=X>

Therefore™ Online:

<https://tenantname.thereforeonline.com/Viewer.aspx?TaskNo=X>

Replace X with the actual number of the task.



Opening a specific Therefore™ Case

Open the Web Viewer first, then redirect to the Case Viewer:

Therefore™ On-Premise:

<http://server/TWA/Viewer.aspx?CaseNo=X>

Therefore™ Online:

<https://tenantname.thereforeonline.com/Viewer.aspx?CaseNo=X>

Open the Case Viewer directly:

Therefore™ On-Premise:

<http://server/TWA/Client/WEB/Viewer/CaseViewer.aspx?CaseNo=X>

Therefore™ Online:

<https://tenantname.thereforeonline.com/Client/WEB/Viewer/CaseViewer.aspx?CaseNo=X>

Replace X with the actual number of the case



Retrieving a workflow instance

Therefore™ On-Premise:

<http://server/TWA/Viewer.aspx?InstanceNo=X>

Therefore™ Online:

<https://tenantname.thereforeonline.com/Viewer.aspx?InstanceNo=X>

Replace X with the instance number.

OR

Therefore™ On-Premise:

<http://server/TWA/TheGetDoc.aspx?ProcessNo=X>

Therefore™ Online:

<https://tenantname.thereforeonline.com/TheGetDoc.aspx?ProcessNo=X>

Replace X with the workflow ID. If the search returns only one item it is automatically opened in the Web Viewer.



Retrieving all workflow instances for a certain process

Therefore™ On-Premise:

<http://server/TWA/Client/thegetdoc.aspx?ProcessNo=all>

Therefore™ Online:

<https://tenantname.thereforeonline.com/Client/thegetdoc.aspx?ProcessNo=all>

If the search returns only one item it is automatically opened in the Web Viewer.



Create new document directly via URL

Therefore™ On-Premise:

<http://server/TWA/TheGetDoc.aspx?CreateNew=true&CtgryNo=X&Index1=Value1&Index2=Value2>

Therefore™ Online:

<https://tenantname.thereforeonline.com/TheGetDoc.aspx?CreateNew=true&CtgryNo=X&Index1=Value1&Index2=Value2>

Replace X with the category number, and Index1, Index2, etc. with the actual column name of the index field. Replace Value1, Value2, etc. with the search values.



Create new case directly via URL

Therefore™ On-Premise:

<http://server/TWA/TheGetDoc.aspx?CreateNew=true&CaseDefNo=X&Index1=Value1&Index2=Value2>

Therefore™ Online:

<https://tenantname.thereforeonline.com/TheGetDoc.aspx?CreateNew=true&CaseDefNo=X&Index1=Value1&Index2=Value2>

Replace X with the case definition number, and Index1, Index2, etc. with the actual column name of the index field. Replace Value1, Value2, etc. with the search values.

5. Therefore™ Portal

5.1 Customization

The Therefore™ Portal can be highly customized to fit the look and feel of the company using it or the website in which it's embedded. The customization is set by editing an XML file on the IIS server. Instructions for editing this file can be found in the Administrator Manual. The following sections outline the finer options available for customization.

5.1.1 Customization File Parameters

The following is a description of the parameters found in the Portal's customization file.

Parameter	Description	Comments
<Skin>	The overall skin style of the Portal	Available values listed as a comment
<PanelWidth>	Width of the embedded Portal	Pixels, independent of iframe width
<FontColor>	Text color in the Portal	Hex value
<LinkColor>	Color of the "Forgot Password" link on the Portal login page and the "Sign Out" link.	Hex value
<InfoFontColor>	The font color for information dialogs.	
<InfoBackgroundColor>	The background color for information dialogs.	
<InfoIcon>	The icon for information dialogs.	
<ConfirmIcon>	The icon for confirmation dialogs.	
<ConfirmFontColor>	The font color for confirmation dialogs.	
<ConfirmBackgroundColor>	The background color for confirmation dialogs.	
<ErrorFontColor>	The font color for error dialogs.	
<ErrorBackgroundColor>	The background color for error dialogs.	
<ErrorIcon>	The icon for error dialogs.	
<DisplayUserName>	Displays the user name in the bottom-right corner.	
<DisplayDisconnectButton>	Displays the "Sign Out" link in the bottom-right corner.	
<NotificationCloseIcon>	The "close window" icon on notification dialogs.	
<ControllerPosition>	The position of the bar where the user name and "Sign Out" link are.	Values: "top" or "bottom"
<ControllerBackgroundColor>	The color of the bar listed above.	Hex value

<LoginTitle>	The text displayed at the top of the login page.	
<LoginUserText>	The text displayed for the user name field.	
<LoginPassword>	The text displayed for the password field.	
<LoginAllowForgotPassword>	Displays the "Forgot Password" link.	
<LoginForgotPasswordText>	The text displayed for the "Forgot Password" link.	
<LoginStaySignedInText>	The text displayed for the "Automatically sign in" checkbox.	
<LoginButtonIcon>	The icon displayed on the "Log on" button.	
<LoginConnectButtonText>	The text displayed on the "Log on" button.	
<NavPanelWidth>	The width of the navigation (left) pane when logged in.	Pixels
<NavQueryIcon>	The icon displayed for searches available in the navigation pane.	
<HitlistPageSize>	The maximum number of hit-list results displayed on each page.	
<HitlistUseColumnWidth>	If set to True, you can set the column width (next parameter).	
<HitlistColumnWidth>	The width of the hit-list columns	Pixels
<HitlistPagingEnabled>	Enables hit-list pagination.	
<HitlistShowHeaderEnabled>	Displays the name of the index data columns above the hit-list.	
<HitlistSortingEnabled>	Allows the hit-list to be sorted.	
<HitlistGroupingEnabled>	Allows the hit-list to be grouped by field.	
<HitlistColumnsReorderEnabled>	Allows the hit-list columns to be reordered.	
<HitlistColumnResizeEnabled>	Allows the hit-list columns to be resized.	
<HitlistRowHoverStyleEnabled>	If enabled, hit-list results are highlighted while hovering over them.	
<HitlistDocumentTypeIconCustomizationEnabled>	If set to True, it will first look in the customization folder for customized document type icons. If one is not found, it will look for the default icons in the Images folder.	Customized type icons must use the following structure: "doctypeicon_xxx.png". Replace xxx with the file extension: i.e. pdf, gif, tif.
<HitlistFilterTextBoxDisplay>	Displays a search box for performing a text-based search.	

<HitlistFilterTextBoxWidth>	The width of the search box listed above.	Pixels
<HitlistFilterTextBoxAlignment>	The alignment of the search box listed above.	Available values: left, center, right
<HitlistFilterTextBoxPosition>	The position of the search box listed above.	Available values: top, bottom
<HitlistOpenButtonDisplay>	Displays the button for opening a document in the hit-list.	
<HitlistOpenButtonColumnWidth>	The width of the column for the "Open" button.	Pixels
<HitlistOpenButtonIcon>	The icon for the "Open" button.	
<HitlistExportButtonDisplay>	Displays the button for exporting a document in the hit-list.	
<HitlistExportButtonColumnWidth>	The width of the column for the "Export" button.	Pixels
<HitlistExportButtonIcon>	The icon for the "Export" button.	
<HitlistIsModifiedDisplay>	Displays the "Modified" column. By default this column is represented by a star symbol. It displays an arrow icon for recently modified documents, or a star icon for newly saved documents.	
<HitlistIsModifiedOnlyNewIcon>	If set to True, only the icon for newly saved documents will be displayed in the "Modified" column.	
<HitlistIsModifiedColumnWidth>	The width of the "Modified" column.	Pixels
<HitlistIsModifiedHeaderIcon>	The header icon for the "Modified" column.	
<HitlistIsModifiedDocumentIcon>	The icon displayed for recently modified documents.	
<HitlistIsModifiedNewDocumentIcon>	The icon displayed for newly saved documents.	
<HitlistDocumentTypeDisplay>	Displays the "Document Type" column. It displays an icon that matches the document format, ie PDF, DOCX, etc.	
<HitlistDocumentTypeColumnWidth>	The width of the "Document Type" column.	Pixels
<HitlistDocumentTypeHeaderIcon>	The header icon for the "Document Type" column.	
<HitlistMultiHitlistExpanded>	Multiple hit-lists, such as when opening a case, are all automatically expanded.	

<HitlistCaseIndexDataColumnDisplay>	Displays the index data of cases in the hit-list. If set to False, only the the next three fields will be displayed. These can also be turned off.	
<HitlistCaseCreatedOnColumnDisplay>	Displays the column "Created On" for cases in the hit-list.	
<HitlistCaseCreatedByColumnDisplay>	Displays the column "Created By" for cases in the hit-list.	
<HitlistCaseSizeColumnDisplay>	Displays the column "Size" for cases in the hit-list.	
<HitlistOnlyCaseAutoOpenEnabled>	If only 1 case is available in the hit-list, it will be automatically expanded without having to click on it.	
<HitlistEmptyMultiResultDisplay>	When opening a case, displays case categories even if they contain no documents.	
<PoweredByFooterAlignment>	Alignment of the "Powered by Therefore™" footer.	Available values: left, center, right. This object is not removable.
<ActivationTitle>	Title of the user activation page.	
<ActivationUser>	The text displayed for the user name field on the activation page.	
<ActivationButton>	The text displayed for the activation button on the activation page.	
<ActivationSuccessfully>	The text displayed for the success message when activating an account.	
<ActivationMailSubject>	The subject of the email sent to users after successful account activation.	
<ActivationMailContent>	The body of the email sent to users after successful account activation.	The mail content must contain the placeholder %s for the password.
<RealTimeHitlist>	Customizes the delay time between key presses for real-time hit-list filtering	

5.1.2 Images

Images used by the customizations can be found in two locations. A default set of images is provided in the TWA\Portal\Images folder. These are the images used by the default customization template. We do not recommend overwriting these. If you want to change an image, copy the new image into your customization folder and change the path in the XML file. As a general recommendation you should keep all the images in the Images folder unchanged so you can reference these for any customization.

Some more images are provided in the "template" customization folder. These are some common alternatives you may find useful. The default template does make use of these.

Document type icons can be displayed in a Portal hit-list, but these are sourced a bit differently. First you need to set the parameters `<HitlistDocumentTypeDisplay>` and `<HitlistDocumentTypeIconCustomizationEnabled>` to TRUE. This will first search the customization folder for customized document type icons. If one is not found, it will look for the default icons in the Images folder. You can add your own customized type icons to your customization folder. These must have the following name structure: "doctypeicon_XXX.png". Replace XXX with the file extension, i.e. pdf, gif, tif.

5.2 Using Multiple Customizations

Multiple customizations can be created for the same Portal. This is useful for centralized systems where one Portal should reflect different branding for different customers/companies. Note that this method only works for single-tenant systems. On multi-tenant systems, each tenant has its own Portal, but a tenant cannot have multiple customizations.

1. Create different customizations in different folders as described in the Administrator Manual.
2. Open IIS Manager and right-click on Default Web Site. Select Add Application.
3. Enter an alias. This needs to be the same name as your customization folder and its XML file.
4. For the Application pool, select Therefore™ Portal.
5. Set the Physical path to your Portal directory, normally C:\inetpub\wwwroot\TWA\Portal.

You can now connect to the Portal with the following URL:

[http://\\$SERVER_NAME/\\$ALIAS/portal.aspx?qList=true](http://$SERVER_NAME/$ALIAS/portal.aspx?qList=true) to make the navigation tree visible to users, which allows them to access different searches. This should be activated if users are assigned more than one search.

OR

[http://\\$SERVER_NAME/\\$ALIAS/portal.aspx?qList=XXX](http://$SERVER_NAME/$ALIAS/portal.aspx?qList=XXX) where XXX is the search ID number. This automatically executes a search, which means the user will immediately see the document hit-list upon logging in. This is useful for scenarios where one search is being used for all users.